

Introduction to neural networks

The perceptron

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1 Data representation

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A dataset is a set of tuples,

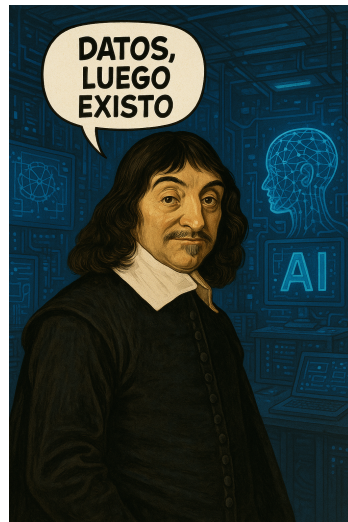
$$\mathcal{D} = \{(X^1, y^1) \dots (X^\mu, y^\mu) \dots (X^m, y^m)\} \quad (1)$$

where X is an input such as an image, and y is the ground truth.

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"Frases que René Descartes nunca dijo".

Dataset implementation

$$\left\{ \begin{bmatrix} \begin{bmatrix} \text{img}_1 \end{bmatrix} \\ \begin{bmatrix} \text{img}_2 \end{bmatrix} \\ \begin{bmatrix} \text{img}_3 \end{bmatrix} \end{bmatrix}, \begin{bmatrix} \text{"gato"} \\ \text{"gato"} \\ \text{"perro"} \end{bmatrix} \right\}$$

$$\mathcal{D} = \left\{ \begin{bmatrix} x_1^1 & \dots & x_n^1 \\ \vdots & \ddots & \vdots \\ x_1^m & \dots & x_n^m \end{bmatrix}, \begin{bmatrix} y^1 \\ \vdots \\ y^m \end{bmatrix} \right\} \quad (2)$$

where for the inputs, X , each row is a example and the columns are features of each example, and for the targets, Y , each element is the target label of that example.

In some occasions, we will write:

$$\mathcal{D} = \{[X^\mu], [Y^\mu]\} \quad (3)$$

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The problem

- We want a machine, f , so that, it says 'yes' when something is detected, namely

$$y \leftarrow f(X) \quad (4)$$



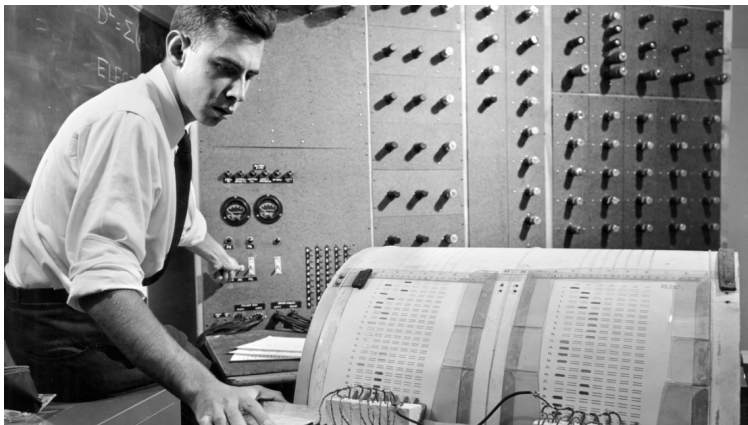


Figure: Frank Rosenblatt '50, Ph.D. '56, works on the “perceptron”¹

¹<https://news.cornell.edu/stories/2019/09/professors-perceptron-paved-way-ai-60-years-too-soon>

Neuron

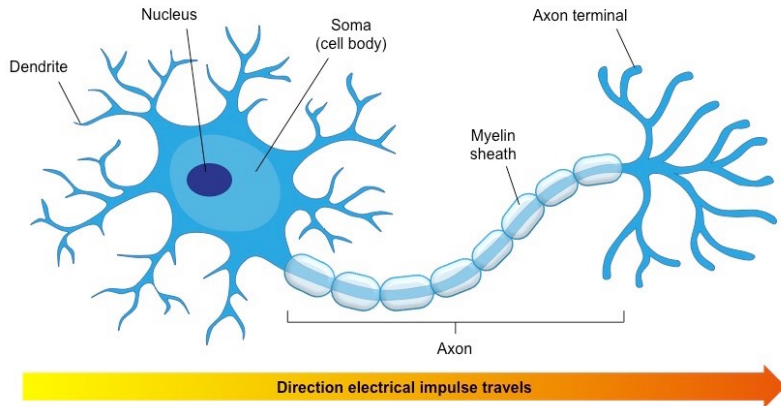


FIG. 1 — Organization of a biological brain. (Red areas indicate active cells, responding to the letter X.)

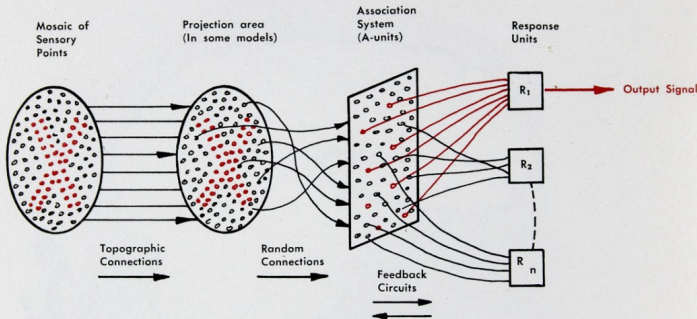


FIG. 2 — Organization of a perceptron.

Figure: "The Design of an Intelligent Automaton," Summer 1958

Perceptron

- “Yet we are about to witness the birth of such a machine – a machine capable of perceiving, recognizing and identifying its surroundings without any human training or control.” Rosenblatt
- “It is a mathematical function mapping some set of input values to output values”

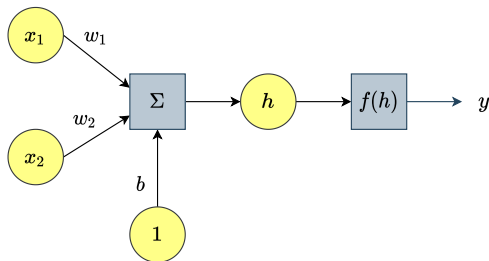


Figure: Perceptron (Extended diagram)

Perceptron

- Weights. Each input to a perceptron has an associated weight that represents its importance.

$$h = \sum_{i=1}^n w_i \cdot x_i + b \quad (5)$$

$$\hat{y} = f(h) \quad (6)$$

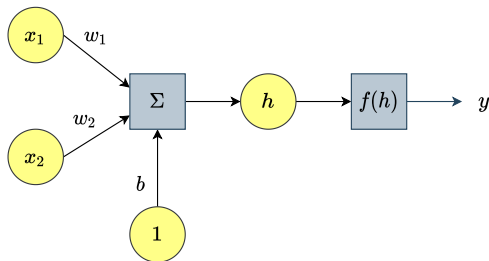
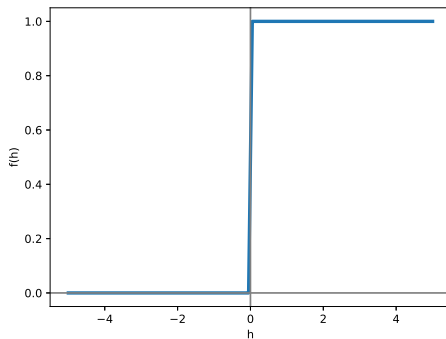


Figure: Perceptron

Perceptron

Activation function

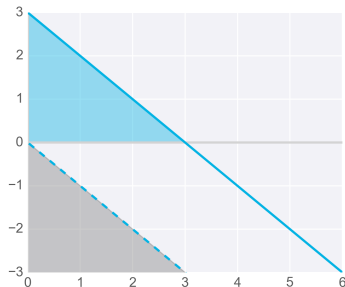
- Activation function $f(h)$.
- Heaviside or step function.



Perceptron

Bias

- One way to get our function to return 1 for more inputs is to add a value to the results of our linear combination, called a bias.



Perceptron

- Full perceptron formula:

$$f(x_1, x_2, \dots, x_n) = \begin{cases} 0 & \text{if } \sum w_i \cdot x_i + b < 0 \\ 1 & \text{if } \sum w_i \cdot x_i + b \geq 0 \end{cases} \quad (7)$$

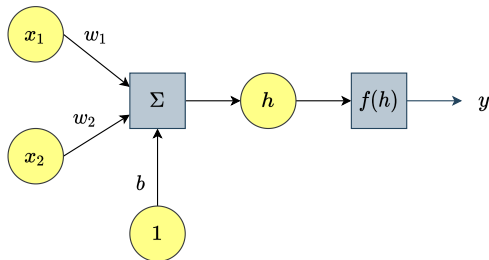


Figure: Perceptron

Comparison against MyP model

- Perceptron (Rosenblatt)

$$f(x_1, \dots, x_n) : \mathbb{R}^n \rightarrow \{0, 1\} \quad (8)$$

- MyP (McCulloch y Pitts)

$$f(x_1, \dots, x_n) : \{0, 1\}^n \rightarrow \{0, 1\} \quad (9)$$

Amount of parameters

The amount of parameters is given by the sum of weights plus biases per layer.

$$\Phi = \{w_1 \dots w_n, b\} \quad (10)$$

$$|\Phi| = n + 1 \quad (11)$$

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Given the perceptron and the targets:

$$Y \approx f(WX + b)? \quad (12)$$

Problem

How do we automatically get the parameters of the perceptron so that the behaviour in a dataset will be replicated?

What to chose?



Figure: Where do we want to go?

What to chose?



Figure: Where do we want to go?

- We want to diminish the total error. We will write it as the loss function.

$$\mathcal{L} = SAE = \sum_{\mu} |y^{\mu} - \hat{y}^{\mu}| \quad (13)$$

- Each error will guide us (left or right):

$$e^{\mu} = y^{\mu} - \hat{y}^{\mu} \quad (14)$$

Perceptron learning rule ²

For a single example:

- Initial step

$$W^0 = [r_1, \dots, r_n] | r_i \sim \mathcal{U}(a, b) \quad (15)$$

- Repeat

$$W^{t+1} = W^t + \eta(y - \hat{y})X, \quad (16)$$

where $\eta > 0$. η is called learning rate. X actuates like a "responsible" of the output.

²Hassoun, Mohamad H. Fundamentals of artificial neural networks. MIT press, 1995.

Perceptron learning rule^{3 4}

For a **dataset** D :

① Initial step

$$W^0 = [r_1, \dots, r_n] | r_i \sim \mathcal{U}(a, b) \quad (17)$$

② Repeat

$$W^{t+1} = W^t + \eta(Y^t - \hat{Y}^t)\mathcal{D}, \quad (18)$$

where $\eta > 0$.

³Rosenblatt, Frank. Principles of neurodynamics: Perceptrons and the theory of brain mechanisms. Vol. 55. Washington, DC: Spartan books, 1962.

⁴Hassoun, Mohamad H. Fundamentals of artificial neural networks. MIT press, 1995.

- Primer equipo exponer: Widrow, Bernard. Adaptive "ADALINE" Neuron Using Chemical memistors. 1960.

References

-  Albert Einstein. *Zur Elektrodynamik bewegter Körper*. (German) [*On the electrodynamics of moving bodies*]. Annalen der Physik, 322(10):891–921, 1905.
-  Knuth: Computers and Typesetting,
<http://www-cs-faculty.stanford.edu/~uno/abcde.html>
-  Joshi, Prateek and Escrivá, David Millán and Godoy, Vinicius, Packt Publishing Ltd, 2016
-  <https://ib.bioninja.com.au/standard-level/topic-6-human-physiology/65-neurons-and-synapses/neurons.html>
-  Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. Probability and statistics for engineers and scientists (9th Ed). New York: Macmillan.